



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC AUTONOMOUS)

B.Tech II Semester Supplementary Examinations June/July-2025

Course Name: ODEs and Multivariable Calculus

(Common for CE, EEE, ME, ECE, CSE, IT, CSC, CSM, CSD, AID & AIM)

Date: 30.06.2025 AN

Time: 3 hours

Max.Marks: 70

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions

Each question carries TWO marks.

10x2=20M

1. Find the solution of $ydx + x(1+y)dy = 0$. 2 M
2. Find the general solution of $y' + y = 0$ 2 M
3. Construct the complementary solution of $(D^2 - 1)y = 8e^{3x}$. 2 M
4. Build the particular integral of $(D^2 + 3D + 3)y = 7e^x$. 2 M
5. Evaluate $\int_3^4 \int_1^2 \frac{dydx}{(xy)^2}$ 2 M
6. Evaluate $\int_0^2 \int_1^z \int_0^{yz} xyz \, dx \, dy \, dz$. 2 M
7. Determine the angle between the surfaces $x \log z = y^2 - 1$ and $x^2y = 2 - z$ at the point $(1, 1, 1)$. 2 M
8. Determine the curl of the vector field $V = (x^2y^2 - z^3)i + 2xyzj + e^{xyz}k$. 2 M
9. Evaluate $\int_C x^2y \, dr$, where C is the curve defined by $x = 3 \cos t, y = 3 \sin t, 0 \leq t \leq \frac{\pi}{2}$. 2 M
10. State Stoke's theorem. 2 M

PART-B

Answer the following. Each question carries TEN Marks.

5x10=50M

- 11.A). i) Construct the solution of $y' + y = y^2$. 5M
 ii) Solve $(1 + y^2) \frac{dx}{dy} = \tan^{-1}y - x$ 5M

OR

- 11.B). Water at temperature $10^\circ C$ takes 5 min to warm up to $20^\circ C$ in a room at temperature $40^\circ C$. (i) Construct the temperature after 20 min. (ii) When will the temperature be $25^\circ C$? 10M

- 12.A). i) Solve $(D^2 - 2D + 10)y = 16e^x \cos 3x$ 5M
 ii) Solve $x^3y''' + 2x^2y'' = x + \sin(\log x)$. 5M

OR

- 12.B). i) Solve $(x - 2)^2y'' - 3(x - 2)y' + 4y = x$. 5M
 ii) Solve $(D^2 + 1)y = \operatorname{cosec} x \cdot \cot x$ by variation of parameters method. 5M

- 13.A). i) Determine the area bounded by the curves $x^2 = y$, $x = y$. 5M
 ii) Determine the volume of the cylinder $x^2 + y^2 = 4, z = 0$ and $z = 4$. 5M

OR

- 13.B). i) Change the order of integration and the evaluate $\int_0^2 \int_1^{e^x} dy \, dx$. 5M
 ii) Determine the volume of the cylinder $x^2 + y^2 + z^2 = 4$ 5M

(P.T.O..)

14. A). i) Find the directional derivative of $f(x, y) = x^2y^3 + xy$ at $(2, 1)$, in the direction of vector $3\vec{i} - 2\vec{j}$. 5M
 ii) Show that $\nabla \times \nabla f = \vec{0}$ for any $f(x, y, z)$. 5M

OR

14. B). i) Show that $\vec{A} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational. Find a scalar function $f(x, y, z)$ such that $\vec{A} = \nabla f$. 5M
 ii) If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = |\vec{r}|$ then show that $\text{div} \left(\frac{\vec{r}}{r^3} \right) = 0$ 5M

15. A). Prove or disprove that Green's theorem is valid for $f(x, y) = e^{-x} \sin y$, $g(x, y) = e^{-x} \cos y$ and C is the square with vertices at $(0, 0), \left(\frac{\pi}{2}, 0\right), \left(\frac{\pi}{2}, \frac{\pi}{2}\right), \left(0, \frac{\pi}{2}\right)$. 10M

OR

15. B). Prove or disprove that divergence theorem is valid for $\iint_S (yz \, dy \, dz + zx \, dz \, dx + xy \, dx \, dy)$, where S is the surface of the cube bounded by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$. 10M
